

FACULTY OF COMPUTING
SEM I - 2024/2025

SECP2523 – DATABASE
ASSIGNMENT 1 – NORMALIZATION

QUESTION 1

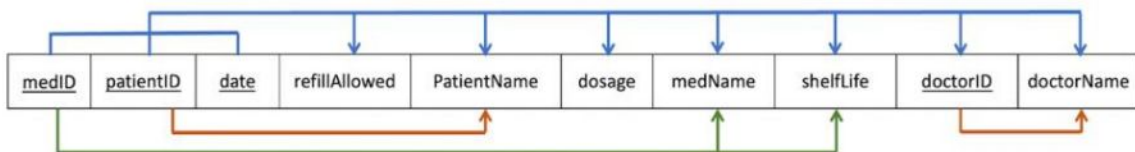


Figure 1: Patient Relation

- Identify **ALL functional dependencies** in the Patient relation (in Figure 1).
- Using the partial dependencies (if they exist in Figure 1), produce a set of relations in Second Normal Form (2NF).
- Using the transitive dependencies (if they exist in Figure 1), produce a set of relations in Third Normal Form (3NF).

QUESTION 2

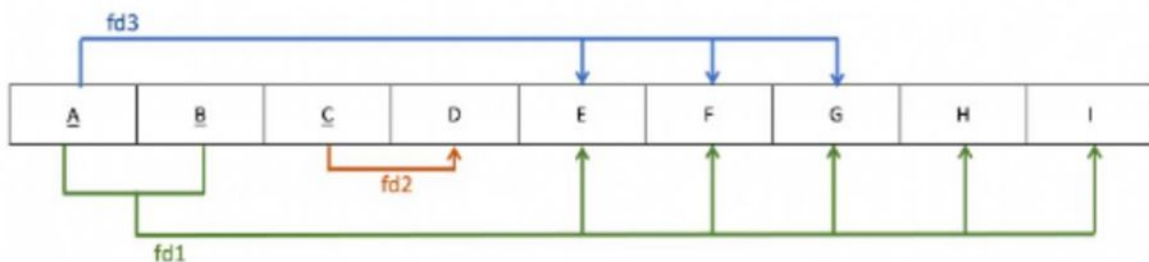


Figure 2: ABC Relation

- Identify **ALL functional dependencies** in the ABC relation (in Figure 2).
- Using the partial dependencies (if they exist in Figure 2), produce a set of relations in Second Normal Form (2NF).
- Using the transitive dependencies (if they exist in Figure 2), produce a set of relations in Third Normal Form (3NF).

QUESTION 3

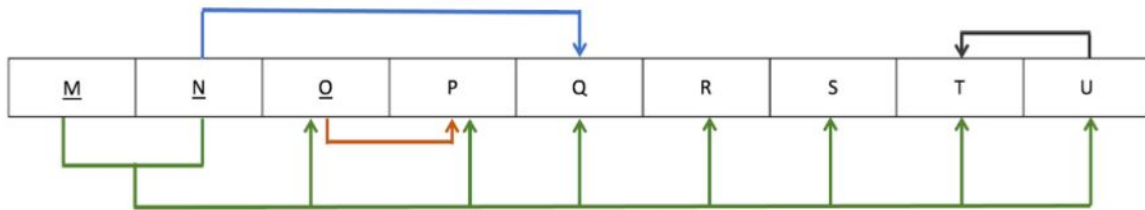


Figure 3: MNQ Relation

- Identify **ALL** functional dependencies in the MNQ relation (in Figure 3).
- Using the partial dependencies (if they exist in Figure 3), produce a set of relations in Second Normal Form (2NF).
- Using the transitive dependencies (if they exist in Figure 3), produce a set of relations in Third Normal Form (3NF).

QUESTION 4

Consider the relation schema LOTS shown in **Table 3**, which describes parcels of land for sale in various counties of a state. Suppose that there are two candidate keys: **PropertyID** and **{County, LotNo}**; that is, **lot numbers** are unique only within each **county**, but **PropertyID** numbers are unique across **counties** for the entire state. **PropertyID** is chosen as the primary key, and it is underlined in **Table 3**.

Table 3: LOTS relation

<u>PropertyID</u>	County	LotNo	Area (Acre)	Price (USD)	TaxRate
P00121	Shawnee	LN02	0.5	500,000	5.5
P05155	Lyon	LN02	1.2	800,000	7.5
P01215	Shawnee	LN21	0.5	500,000	5.5
P20110	Lyon	LN31	1.4	1,000,000	7.5
P50551	Shawnee	LN52	0.6	600,000	5.5
P11111	Shawnee	LN31	0.7	700,000	5.5
P07123	Lyon	LN25	1.4	1,000,000	7.5

- Identify all functional dependencies in the LOTS relation.
- Based on the general definition of normal forms, what normal form is LOTS currently in? Justify your answer.
- Produce a set of relational schemas for the LOTS relation, normalizing it from its current normal form to Third Normal Form (3CNF).

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<u>medID</u>	<u>patientID</u>	<u>date</u>	refillAllowed	PatientName	dosage	medName	shelfLife	<u>doctorID</u>	doctorName
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(a) FD1: medID, patientID, date $\rightarrow$ refillAllowed, PatientName, dosage, medName, shelfLife, doctorName (Primary Key)

FD2: medID $\rightarrow$ medName, shelfLife (Partial Dependency)

FD3: patientID $\rightarrow$ PatientName (Partial Dependency)

FD4: doctorID $\rightarrow$ doctorName (Transitive Dependency)

INF: R (medID, patientID, date, refillAllowed, PatientName, dosage, medName, shelfLife, doctorID, doctorName)

(b) 2NF

From FD2: Medication (medID, medName, shelfLife)
PK: medID

PK: medID, patientID, date

From FD3: Patient (patientID, PatientName)
PK: patientID

The rest: Prescription (medID, patientID, date, refillAllowed, dosage, doctorID, doctorName)
PK: medID, patientID, date

FK: medID reference Medication (medID)

FK: patientID reference Patient (patientID)

(c) 3NF

From FD2 : Medication (medID , medName , shelfLife)
PK: medID

From FD3 : Patient (patientID , PatientName)
PK: patientID

From FD4 : Doctor (doctorID , doctorName)
PK: doctorID

The rest : Prescription (medID , patientID , date , refillAllowed , dosage , doctorID)
PK: medID , patientID , date

FK: medID reference Medication (medID)

FK: patientID reference Patient (patientID)

FK: doctorID reference Doctor (doctorID)

2-(a) FD1 : $A, B, C \rightarrow D, E, F, G, H, I$ (primary key)
FD2 : $C \rightarrow D$ (partial dependency)
FD3 : $A \rightarrow E, F, G$ (partial dependency)

INF : $R(\underline{A}, \underline{B}, \underline{C}, D, E, F, G, H, I)$
PK: A, B, C

(b) 2NF

From FD2 : $R_1(\underline{C}, D)$
PK: C

From FD3 : $R_2(\underline{A}, E, F, G)$
PK: A

The Rest : $R(\underline{A}, \underline{B}, \underline{C}, H, I)$
PK: A, B, C

FK: A reference (R2) A

FK: C reference (R1) C

(c) 3NF

No transitive dependency

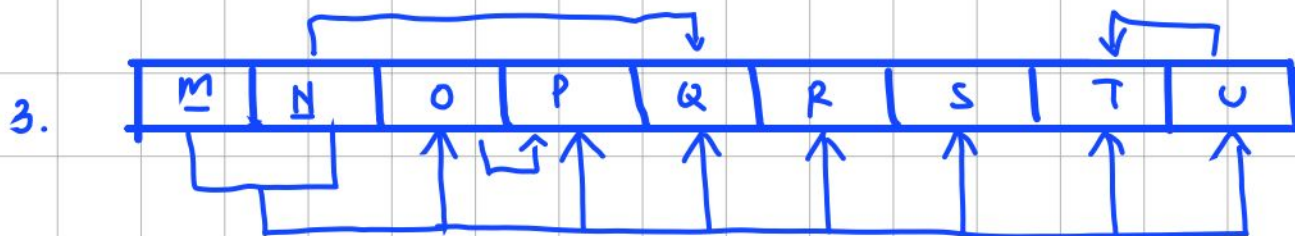
From FD2 : $R_1(\underline{C}, D)$
PK: C

From FD3 : $R_2(\underline{A}, E, F, G)$
PK: A

The Rest : $R(\underline{A}, \underline{B}, C, H, I)$
PK: A, B, C

FK: A reference (R2) A

FK: C reference (R1) C



(a) FD 1: $M, N \rightarrow O, P, Q, R, S, T, U$ (primary key)

FD 2: $O \rightarrow P$ (transitive dependency)

FD 3: $N \rightarrow Q$ (partial dependency)

FD 4: $U \rightarrow T$ (transitive dependency)

(b) FD 3: $R_1(\underline{N}, Q)$ PK: N

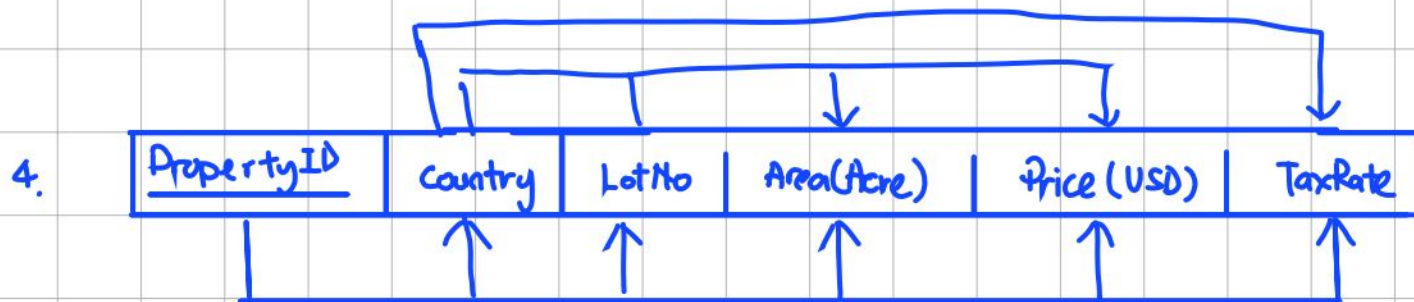
Rest: $R(\underline{M}, \underline{N}, O, P, R, S, T, U)$ PK: M, N ,
FK: N reference $R_1(N)$

(c) FD 3: $R_1(\underline{N}, Q)$ PK: N

FD 2: $R_2(\underline{O}, P)$ PK: O

FD 4: $R_3(\underline{U}, T)$ PK: U

Rest: $R(\underline{M}, \underline{N}, O, R, S, U)$ PK: M, N
FK: N reference $R_1(N)$
FK: U reference $R_3(U)$
FK: O reference $R_2(O)$



(a) FD1: $\text{PropertyID} \rightarrow \text{Country}, \text{LotNo}, \text{Area(Acre)}, \text{Price(USD)}, \text{TaxRate}$ (Primary key)

FD2: $\{\text{Country}, \text{LotNo}\} \rightarrow \text{Area}, \text{Price}$ (candidate key)

FD3: $\text{Country} \rightarrow \text{TaxRate}$ (transitive dependency)

(b) The LOTS relation is in 3NF because PropertyID relates with Country in R1 while Country relates with TaxRate in R3. Every non-primary key attribute depends directly on candidate key. There is transitive dependency between PropertyID, Country and TaxRate. The transitive dependency is $\text{Country} \rightarrow \text{TaxRate}$.

(c) 1NF: R1 (PropertyID, Country, LotNo, Area, Price, TaxRate)
PK: PropertyID.

* No partial dependency

2NF: R1 (PropertyID, Country, LotNo, Area, Price, TaxRate)
PK: PropertyID.

3NF: R1 (PropertyID, Country, LotNo, Area, Price)
PK: PropertyID.

FK: Country reference R2(Country)

FK: LotNo reference R2(LotNo)

FK: Country reference R3(Country)

R2(Country , LotNo , Area, Price)

PK: Country, LotNo

R3(Country , TaxRate)

PK: Country